# Slime Mold Simulation: Theory & Data Analysis

## 1 Theoretical Derivation of Classical Keller-Segel

### 1.1 Diffusion Equation

We start from the continuity equation,

$$\frac{\partial}{\partial t}\int_V C\,dV = -\int_S \vec{j}\cdot d\vec{\sigma} + \int_V f\,dV \tag{1}$$

where $C$ is the concentration, $\vec{j}$ is the flux, and $f$ is the source term. Using the divergence theorem, we get the conservation equation,

$$\int_V \left[\frac{\partial C}{\partial t} + \nabla\cdot\vec{j} - f\right] dV = 0 \implies \frac{\partial C}{\partial t} + \nabla\cdot\vec{j} = f. \tag{2}$$

Using Fick's law of diffusion, the flux is proportional to the gradient of the field: $\vec{j} = -D\nabla C$. Thus, we get

$$\frac{\partial C}{\partial t} = f + \nabla\cdot(D\nabla C). \tag{3}$$

In our specific model for chemical concentration $c$, we get

$$\frac{\partial c}{\partial t} = D_{\text{chem}}\nabla^2 c - \gamma c + \alpha\rho \tag{4}$$

with diffusion constant $D_{\text{chem}}$, degradation rate $\gamma$ and secretion rate $\alpha$ proportional to cell density $\rho$.

### 1.2 Chemotaxis

We use a continuous model for cell concentration here. Let's suppose the presence of a chemical concentration gives rise to the movement of cells. The flux of cells is given by:

$$\vec{J} = \vec{J}_{\text{diff}} + \vec{J}_{\text{chemo}} \tag{5}$$

where $\vec{J}_{\text{diff}} = -D_{\text{cell}}\nabla\rho$ and the chemotactic flux is $\vec{J}_{\text{chemo}} = \rho\chi(c)\nabla c$. Here, $\chi(c)$ is the chemotactic sensitivity function.

The cell population also follows the continuity equation

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{J} = f(\rho). \tag{6}$$

Substituting the flux terms gives

$$\frac{\partial \rho}{\partial t} = f(\rho) + \nabla \cdot (D_{\text{cell}} \nabla \rho) - \nabla \cdot (\rho \chi(c) \nabla c). \tag{7}$$

For constant parameters and no cell growth or production ($f(\rho) = 0$), the equation is simplified to

$$\frac{\partial \rho}{\partial t} = D_{\text{cell}} \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c). \tag{8}$$

In a more realistic model, chemotactic sensitivity function is usually modelled as $\chi(\rho, c) = \frac{b\rho}{(a+c)^2}$, where $a$ and $b$ are parameters to be determined. For simplification, $\chi$ is set to be constant in this simulation.

**Summary of Equations:**

1. $$\frac{\partial c}{\partial t} = D_{\text{chem}} \nabla^2 c - \gamma c + \alpha \rho \tag{9}$$
2. $$\frac{\partial \rho}{\partial t} = D_{\text{cell}} \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c) \tag{10}$$

Generally, diffusion is a stabilizing force while chemotaxis is destabilizing. The balance between stabilization and destabilization can result in steady states or unsteady, wave-like patterns. If chemotaxis is too strong, the solution blows up.

### 1.3 Linear Stability Analysis for the Keller-Segel Model

To find the collapse threshold, we check if the uniform distribution is stable against tiny fluctuations:

$$\rho(r, t) = \rho_0 + \delta\rho \quad \text{and} \quad c(r, t) = c_0 + \delta c \tag{11}$$

Plugging these into Equation 8 and ignoring the second-order term $\delta\rho \nabla \delta c$:

$$\frac{\partial \delta\rho}{\partial t} = D_{\text{cell}} \nabla^2 \delta\rho - \chi \rho_0 \nabla^2 \delta c \tag{12}$$

For the chemical field (Equation 4), we assume a quasi-steady state that the chemical diffuses much faster than the cells ($\frac{\partial c}{\partial t} \approx 0$). Thus we get

$$0 = D_{\text{chem}} \nabla^2 \delta c - \gamma \delta c + \alpha \delta \rho. \tag{13}$$

By doing this, we linearized the two PDEs! Now we can solve Equation 12 and 13 using usual method (Fourier transform for spatial domain and Laplace transform for the time domain which is initial value problem).
Rules:

- FT: $\mathcal{F}(\nabla^2 f) = -k^2 \tilde{f}$
- LT: $\mathcal{L}\left(\frac{df}{dt}\right) = \omega \tilde{f} - f(t = 0)$

Transforming two equations gives

$$\omega\delta\tilde{\rho} - \delta\tilde{\rho}_i = -k^2(D_{\text{cell}}\delta\tilde{\rho} - \chi\rho_0\delta\tilde{c}) \tag{14}$$

$$0 = -D_{\text{chem}}k^2\delta\tilde{c} - \gamma\delta\tilde{c} + \alpha\delta\tilde{\rho} \tag{15}$$

where $\delta\rho_i = \delta\rho(t=0)$. We can solve them as $2 \times 2$ matrix problem

$$\begin{pmatrix} \omega + D_{\text{cell}}k^2 & -\chi\rho_0 k^2 \\ -\alpha & \gamma + D_{\text{chem}}k^2 \end{pmatrix} \begin{pmatrix} \delta\tilde{\rho} \\ \delta\tilde{c} \end{pmatrix} = \begin{pmatrix} \delta\tilde{\rho}_i \\ 0 \end{pmatrix}. \tag{16}$$

To solve for $\delta\tilde{\rho}$, invert the matrix and we get

$$\delta\tilde{\rho}(k,\omega) = \frac{\delta\tilde{\rho}_i}{\omega + D_{\text{cell}}k^2 - \frac{\chi\rho_0\alpha k^2}{\gamma + D_{\text{chem}}k^2}}. \tag{17}$$

Using the inverse Laplace transform $\mathcal{L}^{-1}\left(\frac{1}{s-a}\right) = e^{at}$, the time-domain solution is

$$\delta\tilde{\rho}(k,t) = \delta\tilde{\rho}_i(k)e^{\omega(k)t} \tag{18}$$

where the dispersion relation is obtained as

$$\omega(k) = -D_{\text{cell}}k^2 + \frac{\chi\rho_0\alpha k^2}{\gamma + D_{\text{chem}}k^2}. \tag{19}$$

### 1.4 Dispersion Relation and Critical Density

- When $\omega < 0 \implies$ exponential decay (stable mode).
- When $\omega > 0 \implies$ exponential growth (unstable mode $\implies$ collapse condition).

Setting $\omega(k) > 0$:

$$\left(\frac{\chi\rho_0\alpha}{\gamma + D_{\text{chem}}k^2} - D_{\text{cell}}\right)k^2 > 0, \tag{20}$$

assuming $k \neq 0$, the minimum $\rho_0$ for instability emerges when $k \to 0$, so we get

$$\rho_0 > \frac{D_{\text{cell}}(\gamma + D_{\text{chem}}k^2)}{\chi\alpha} \implies \rho_c = \frac{D_{\text{cell}}\gamma}{\chi\alpha} \tag{21}$$

which is the critical density for collapse.

### 1.5 Find the Cluster Separation

To find the cluster separation, we need the fastest growing $k$-mode that overtakes the other modes. We find this by maximizing the dispersion relation $\frac{d\omega}{dk^2} = 0$:

$$-D_{\text{cell}} + \frac{\chi\rho_0\alpha(\gamma + D_{\text{chem}}k^2) - \chi\rho_0\alpha k^2 D_{\text{chem}}}{(\gamma + D_{\text{chem}}k^2)^2} = 0 \tag{22}$$

This simplifies to:

$$D_{\text{cell}}(\gamma + D_{\text{chem}}k^2)^2 = \chi\rho_0\alpha\gamma \implies D_{\text{chem}}k^2 + \gamma = \sqrt{\frac{\chi\rho_0\alpha\gamma}{D_{\text{cell}}}} \tag{23}$$

Solving for $k^2_{\text{max}}$:

$$k^2_{\text{max}} = \frac{1}{D_{\text{chem}}}\left(\sqrt{\frac{\chi\rho_0\alpha\gamma}{D_{\text{cell}}}} - \gamma\right) \tag{24}$$

Using the critical density $\rho_c = \frac{D_{\text{cell}}\gamma}{\chi\alpha}$, the maximum wavelength is

$$\lambda_{\text{max}} = 2\pi\sqrt{\frac{D_{\text{chem}}}{\gamma}}\left(\sqrt{\frac{\rho_0}{\rho_c}} - 1\right)^{-1/2} = 2\pi\lambda\left(\sqrt{\frac{\rho_0}{\rho_c}} - 1\right)^{-1/2} \tag{25}$$

where $\lambda = \sqrt{\frac{D_{\text{chem}}}{\gamma}}$ is the screening length.

### 1.6 Steady-state Density Profile

Starting from the general equation at steady state ($\frac{\partial\rho}{\partial t} = 0$):

$$\nabla \cdot (D_{\text{cell}}\nabla\rho - \chi\rho\nabla c) = 0 \implies D_{\text{cell}}\nabla\rho - \chi\rho\nabla c = 0 \tag{26}$$

Integrating spatially gives

$$\frac{D_{\text{cell}}}{\chi}\frac{\nabla\rho}{\rho} = \nabla c \implies \nabla\left(\frac{D_{\text{cell}}}{\chi}\ln\rho - c\right) = 0 \tag{27}$$

$$\frac{D_{\text{cell}}}{\chi}\ln\rho = c + A \implies \rho(r) = \rho_{bg}\exp\left(\frac{\chi c(r)}{D_{\text{cell}}}\right). \tag{28}$$

This is exactly the Boltzmann distribution where $U(r) = -\chi c(r)$ and $k_BT = D_{\text{cell}}$. This equation doesn't give the actual distribution, but shows the relation between $\rho(r)$ and $c(r)$.

## 2 Data Analysis

### 2.1 Phases of the Evolution of the System

- **Phase 1: Linear phase.** This is the phase where we can use linear perturbation theory. When the simulation first starts, the agents are roughly uniform and $k_{\text{max}}$ dictates the system.
- **Phase 2: Non-linear phase.** The linear approximation breaks down completely. Each cluster behaves like a gravitational well acting with long-range forces.

### 2.2 Finding the Separation Between Clusters

The best way to do this is working in Fourier space. Essentially, I find the power spectrum of the cell distribution to find the peak mode. The problem is: which time step am I supposed to look at to pick the maximum $k$-mode? I need to determine when the linear regime occurs. The best way is to use $\rho_{\text{max}} < 1.1\rho_0$ as a threshold for the linear regime.

### 2.3 Issue with the Initial Setup

The number of cells per pixel follows a binomial distribution:

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k} \quad \text{where } p = \frac{1}{\text{total pixels}} \tag{29}$$

In the limit of large $N$, this becomes a Poisson distribution with mean $\lambda = N \times p = \rho_0$, simply meaning our grids are small enough to treat them as smooth continuous space where the variance is $\sigma^2 = \rho_0$.

The percentage fluctuation is

$$\frac{\delta\rho}{\rho_0} = \frac{\sigma}{\rho_0} = \frac{1}{\sqrt{\rho_0}} \tag{30}$$

Thus if we set $\rho_0$ too small (e.g., 10 per pixel), the noise is too large to ever be in the linear regime. We want $\frac{\delta\rho}{\rho_0} \approx 0.03 \implies \rho_0 = 1000$.

### 2.4 Calculating Physical Length

The discrete Fourier transform is given by:

$$\mathcal{F}(k_x, k_y) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} \delta\rho(x,y) e^{-i2\pi(k_x \frac{x}{N_x} + k_y \frac{y}{N_y})} \tag{31}$$

In continuous space, $\vec{k'} = 2\pi \left( \frac{1}{\lambda_x} \hat{i} + \frac{1}{\lambda_y} \hat{j} \right)$. Thus we have

$$2\pi \sqrt{\frac{k_x^2}{N_x^2} + \frac{k_y^2}{N_y^2}} = |\vec{k'}| = \frac{2\pi}{\lambda} \tag{32}$$

If we assume a square grid $N_x = N_y = N$, then

$$\frac{1}{N}\sqrt{k_x^2 + k_y^2} = \frac{k}{N} = \frac{1}{\lambda} \implies \lambda_{\max} = \frac{N}{k} \quad \text{and} \quad \delta\lambda_{\max} = \lambda_{\max} \frac{\delta k_{\max}}{k_{\max}}. \tag{33}$$

From the context of the simulation parameters: $D_{\text{chem}} = 1$, $\gamma = 0.1$, $\alpha = 1$, $\chi = 2$, $D_{\text{cell}} = D_r = 0.5$.

$$\lambda_{\max} = 2\pi \sqrt{\frac{1}{0.1}} \left( \sqrt{10000 \times \frac{2 \times 1}{0.5 \times 0.1}} - 1 \right)^{-1/2} = 0.79 \tag{34}$$

There is a problem here. $\lambda_{\max} = 0.79$ but $dx = 1$, which means we can't resolve the wavelength. Theoretically, we must have at least $\lambda = 2$ (two pixels) to see a peak and trough. It's safe to explore from $\lambda \sim 3$. Solutions:

- Keep biological parameters and increase resolution of the simulation.
- Keep the resolution but move biological parameters.

### 2.5 Using MLE to Extract $k_{\max}$

To use Maximum Likelihood Estimation (MLE), we need to know the likelihood function for a single data point.

The complex coefficient from the Fourier transform is

$$Z(\vec{k}) = \int \delta\rho(\vec{r}) e^{i\vec{k}\cdot\vec{r}} d\vec{r} = X + iY. \tag{35}$$

By the Central Limit Theorem, we can assume $Z$ is randomly distributed that

$$X, Y \sim \mathcal{N}(0, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{X^2 \text{ or } Y^2}{2\sigma^2}\right). \tag{36}$$

Therefore, the joint probability of getting $X$ and $Y$ is

$$f(X, Y)\, dX\, dY = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{X^2 + Y^2}{2\sigma^2}\right) dX\, dY \tag{37}$$

By doing a coordinate transformation $(X, Y) \to (P, \theta)$ where $P = X^2 + Y^2$ and $dX\, dY = \frac{1}{2} dP\, d\theta$. Integrating over the phase $\theta$ gives

$$f(P)\, dP = \frac{1}{2\sigma^2} \exp\left(-\frac{P}{2\sigma^2}\right) dP \tag{38}$$

which is exponential distribution.

But what is $\sigma^2$? From previous relation, $\sigma^2 = \langle X^2 \rangle = \langle Y^2 \rangle$, thus we can say

$$\sigma^2 = \frac{1}{2}\langle X^2 + Y^2 \rangle = \frac{1}{2} S(k) \tag{39}$$

where $S(k) = \langle P(k) \rangle$, the expected power for given $k$. Here We model $S(k)$ as a Gaussian curve

$$S(k) = A \exp(-\frac{(k - k_{\max})^2}{2w^2}) + C \tag{40}$$

where $A, k, w, C$ are adjustable parameters. So our likelihood function can be written as

$$f(P|S(k)) = \frac{1}{S} \exp\left(-\frac{P}{S}\right). \tag{41}$$

Then log-likelihood function of getting a set of data points becomes

$$\ln \mathcal{L} \propto \sum_i^N \left[-\ln S(k_i) - \frac{P_i}{S(k_i)}\right]. \tag{42}$$

We maximize this function to find $k_{\max}$.

**Important facts:**

1. For a given $k$, the most probable data point is actually $P = 0$. 2. Trying to fit the Gaussian curve by binning the data points with symmetric Gaussian error can lead to significant flaw as the distribution of the data points are highly skewed.

**Extra derivation: Why can we assume $\sigma_{X^2} = \sigma_{Y^2}$**
In short, it's because the system is translationally symmetric.

$$X = \int \delta\rho(\vec{r}) \cos(\vec{k} \cdot \vec{r}) d\vec{r}, \qquad Y = \int \delta\rho(\vec{r}) \sin(\vec{k} \cdot \vec{r}) d\vec{r} \tag{43}$$

then

$$\sigma_X^2 = \langle X^2 \rangle = \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot \vec{r}) \cos(\vec{k} \cdot \vec{r}') d\vec{r} d\vec{r}' \tag{44}$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \left[ \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) + \cos(\vec{k} \cdot (\vec{r} + \vec{r}')) \right] d\vec{r} d\vec{r}' \tag{45}$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) d\vec{r} d\vec{r}' \tag{46}$$

as our system must be independent on the absolute coordinates $\vec{r} + \vec{r}'$ and only dependent on the relative coordinates $\vec{r} - \vec{r}'$. Similarly

$$\sigma_Y^2 = \langle Y^2 \rangle = \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \sin(\vec{k} \cdot \vec{r}) \sin(\vec{k} \cdot \vec{r}') d\vec{r} d\vec{r}' \tag{47}$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \left[ \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) - \cos(\vec{k} \cdot (\vec{r} + \vec{r}')) \right] d\vec{r} d\vec{r}' \tag{48}$$

$$= \frac{1}{2} \iint \delta\rho(\vec{r}) \delta\rho(\vec{r}') \cos(\vec{k} \cdot (\vec{r} - \vec{r}')) d\vec{r} d\vec{r}' \tag{49}$$

Thus $\sigma_X^2 = \sigma_Y^2$

## 2.6 Error Estimation of the MLE Fit

To estimate the errors for our Maximum Likelihood Estimation, we evaluate the Fisher Information Matrix. For our specific MLE, the log-likelihood function is:

$$\ln \mathcal{L}(P|\theta) = \sum_{i=1}^{N} \left( -\ln S_i(\theta) - \frac{P_i}{S_i(\theta)} \right) \tag{50}$$

The first derivative with respect to a parameter $\theta_j$ is:

$$\frac{\partial \ln \mathcal{L}_i}{\partial \theta_j} = -\frac{1}{S_i} \frac{\partial S_i}{\partial \theta_j} + \frac{P_i}{S_i^2} \frac{\partial S_i}{\partial \theta_j} = \frac{\partial S_i}{\partial \theta_j} \left( -\frac{1}{S_i} + \frac{P_i}{S_i^2} \right). \tag{51}$$

The second derivative (the Hessian matrix $H_{i,jk}$) is calculated as:

$$H_{i,jk} = \frac{\partial^2 \ln \mathcal{L}_i}{\partial \theta_j \partial \theta_k} = \frac{\partial^2 S_i}{\partial \theta_j \partial \theta_k} \left( -\frac{1}{S_i} + \frac{P_i}{S_i^2} \right) + \frac{\partial S_i}{\partial \theta_j} \frac{\partial S_i}{\partial \theta_k} \left( \frac{1}{S_i^2} - \frac{2P_i}{S_i^3} \right). \tag{52}$$

By taking the expectation value and using the relation $\langle P_i \rangle = S_i$, the first term vanishes:

$$\langle H_{i,jk} \rangle = \frac{\partial^2 S_i}{\partial\theta_j \partial\theta_k}\left(-\frac{1}{S_i} + \frac{S_i}{S_i^2}\right) + \frac{\partial S_i}{\partial\theta_j}\frac{\partial S_i}{\partial\theta_k}\left(\frac{1}{S_i^2} - \frac{2S_i}{S_i^3}\right) = -\frac{1}{S_i^2}\frac{\partial S_i}{\partial\theta_j}\frac{\partial S_i}{\partial\theta_k}. \tag{53}$$

The joint expected Hessian over all data points becomes:

$$\langle H_{jk} \rangle = \sum_i \langle H_{i,jk} \rangle = -\sum_i \frac{1}{S_i^2}\frac{\partial S_i}{\partial\theta_j}\frac{\partial S_i}{\partial\theta_k}. \tag{54}$$

The covariance matrix $\Sigma$ is the inverse of the Fisher Information Matrix $\mathcal{I}$:

$$\Sigma = \mathcal{I}^{-1} = \left(\sum_i \frac{1}{S_i^2}\frac{\partial S_i}{\partial\theta_j}\frac{\partial S_i}{\partial\theta_k}\right)^{-1}. \tag{55}$$

**Key point**:
When we have enough data, Hessian matrix converges to expected Hessian. Thus instead of calculating the 2nd derivatives term, we can directly use theoretical expected Hessian which is computationally cheaper.

To compute this efficiently, we define the Jacobian matrix $J_{ij} = \frac{\partial S_i}{\partial\theta_j}$ and a diagonal weight matrix $W_{ii} = \frac{1}{S_i^2}$. This allows us to write the Information Matrix as:

$$\mathcal{I}_{jk} = \sum_i W_{ii} J_{ij} J_{ik} = (J^T W J)_{jk} \implies \mathcal{I} = J^T W J. \tag{56}$$

For our expected power model $S_i(f_i | f_{\max}, A, w, C) = A\exp\left(-\frac{(f_i - f_{\max})^2}{2w^2}\right) + C$, the partial derivatives are:

$$\frac{\partial S_i}{\partial A} = \exp\left(-\frac{(f_i - f_{\max})^2}{2w^2}\right) \tag{57}$$

$$\frac{\partial S_i}{\partial w} = A\exp\left(-\frac{(f_i - f_{\max})^2}{2w^2}\right)\frac{(f_i - f_{\max})^2}{w^3} \tag{58}$$

$$\frac{\partial S_i}{\partial f_{\max}} = A\exp\left(-\frac{(f_i - f_{\max})^2}{2w^2}\right)\frac{f_i - f_{\max}}{w^2} \tag{59}$$

$$\frac{\partial S_i}{\partial C} = 1. \tag{60}$$

We construct the Jacobian $J$ by column-stacking these partial derivatives. To optimize the computation, instead of full matrix multiplication with the diagonal $W$, we perform a Hadamard product between the broadcasted weight vector and $J$ before multiplying by $J^T$.

# 3 Results

## 3.1 23/09/2026

Not sure what is exactly the problem at this point. Two possible reason why my simulation failed to reproduce the theoretical relation.

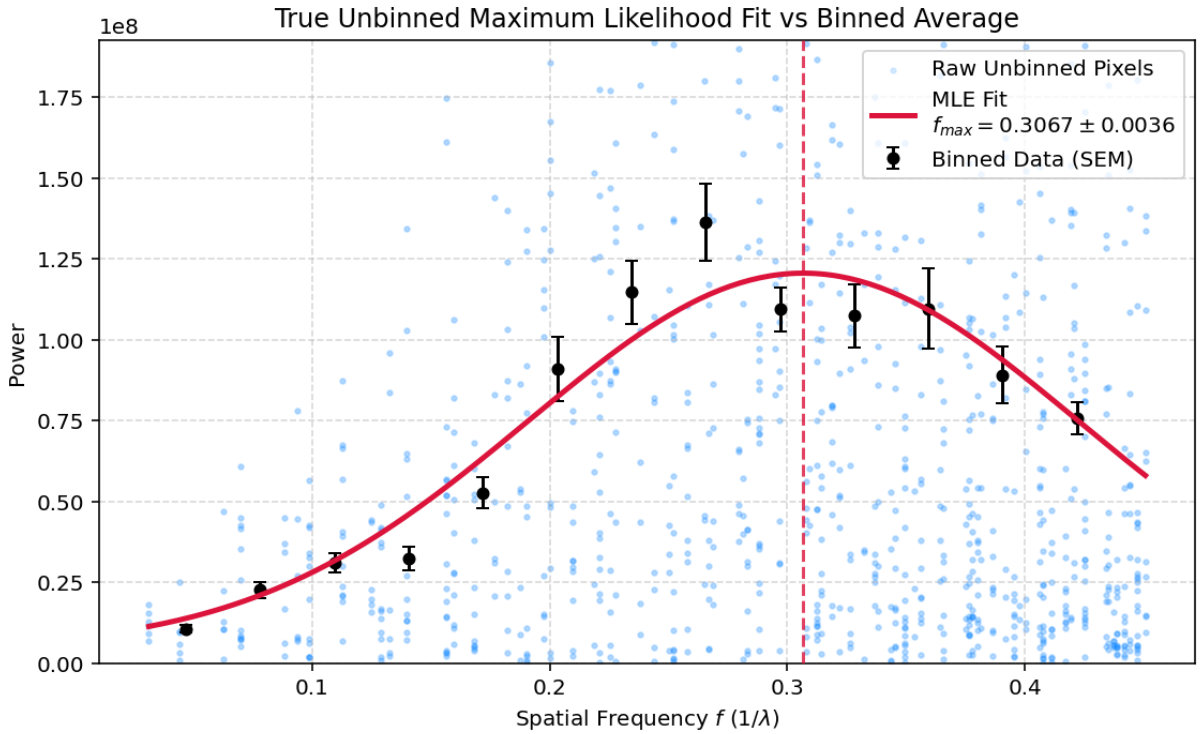



Figure 1: This is example of MLE fitting for Dchem = 16.

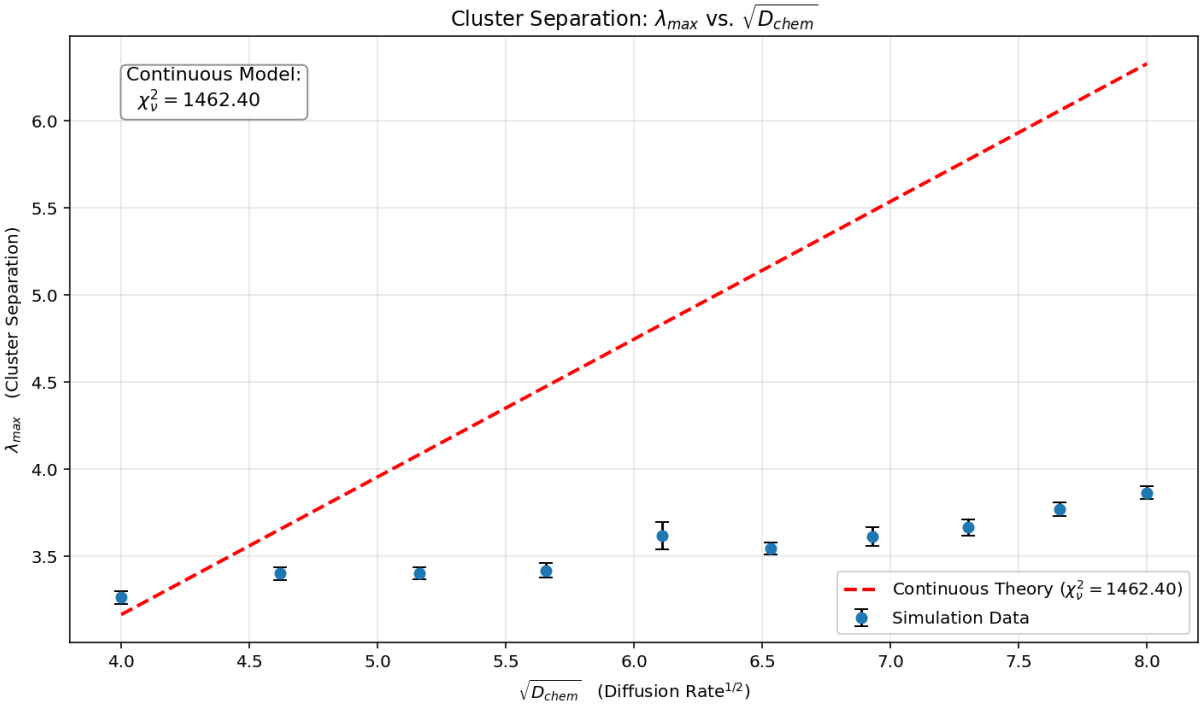



Figure 2: Also the final results clearly do not follow the theory. I guess the low resolution of the simulation is the problem, but I am not sure.

1. Wrong setup for initial chemical field. It must be $c_0 = \frac{\alpha}{\gamma}\rho_0$ but currently $c_0 = 0$. By the way I shouldn't introduce $\delta c$ in the initial setup because $\frac{\partial c}{\partial c} \approx 0$

2. $D_{\text{chem}}$ is too small to satisfy $\frac{\partial c}{\partial t} \approx 0$ Here are several ways to solve this problem.
a. Inflate $D_{\text{chem}}$, but in this case the $\lambda_{\max}$ can exceed the grid size. So after all I need to increase the resolution. Not suitable.
b. Substep the chemical field. This makes sure the chemical field is always in steady state against the cell density. However it costs a lot computationally as well as it increases the total number of calculation.
c. Inflate dt for chemical field. This is the most straight forward methods. Computationally, it costs 0. Just need to be careful that the simulation satisfies Von Neumann stability limits that

$$D_{chem}\frac{dt}{dx^2} \leq \frac{1}{4} \tag{61}$$

Current dt = 0.00001 and dx = 1. So we can safely inflate dt upto 300 times 0.00001 to satisfy the limit for $D_{\text{chem}} = 64$

### 3.2 25/09/2026

**Re-examining the Quasi-Steady State Assumption**
From the linear stability analysis, if we do not assume $\frac{\partial c}{\partial t} \approx 0$, the perturbed chemical equation is:

$$\frac{\partial \delta c}{\partial t} = D_{\text{chem}}\nabla^2\delta c - \gamma\delta c + \alpha\delta\rho. \tag{62}$$

Transforming this into Fourier space yields a first-order ODE with a source term:

$$\frac{\partial \delta\tilde{c}}{\partial t} = -(D_{\text{chem}}k^2 + \gamma)\delta\tilde{c} + \alpha\delta\tilde{\rho}. \tag{63}$$

The homogeneous solution governing the relaxation of the chemical field is $\delta\tilde{c} = Ce^{-\Gamma_c t}$, where the chemical relaxation rate is $\Gamma_c = D_{\text{chem}}k^2 + \gamma$. For the quasi-steady state assumption $\frac{\partial c}{\partial t} \approx 0$ to be valid, the relaxation rate of the chemical field ($\Gamma_c$) must be significantly greater than the growth rate of the cell clusters ($\omega$). However, with our current configuration, the ratio $\omega/\Gamma_c$ is approximately 20 for $D_{\text{chem}} = 16$ and approximately 5 for $D_{\text{chem}} = 64$. Because this ratio is high, the cells are clustering much faster than the chemical field can relax.

To satisfy the assumption, we must artificially increase the relaxation rate of the chemical field. We can achieve this by setting an inflated time step $dt_{\text{chem}} = M \times dt$ with a multiplier of $M = 300$. We must check if this maintains numerical stability via the Von Neumann limit:

$$D_{\text{chem}}\frac{dt_{\text{chem}}}{dx^2} = 64 \times \frac{0.00001 \times 300}{1^2} = 0.192 \leq 0.25. \tag{64}$$

This confirms that the inflated time step satisfies the numerical stability limit while successfully decoupling the timescales.

### 3.3 26/06/2026

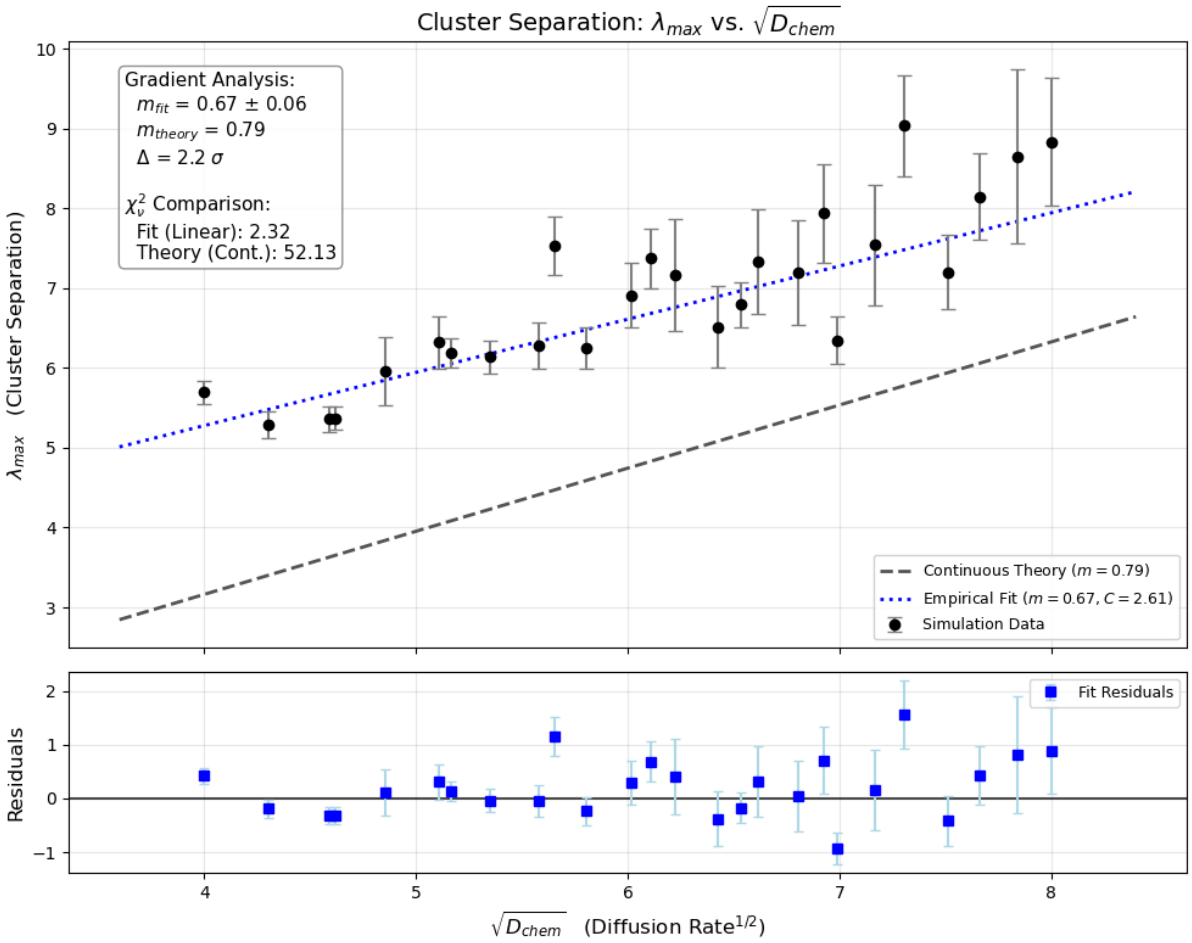



Figure 3: linear model y = mx + C with m = 0.67 and C = 2.61

The simulation with artificially inflated time-step for chemical field produced interesting results.

1. It correctly reproduced the relationship $\lambda \propto \sqrt{D_{\text{chem}}}$ with the gradient $m = 0.67 \pm 0.06$ which is 2 standard deviation away from the theoretical gradient

2. It has offset C = 2.6. This actually make sense. In this simulation, the pixelated space prevents system from having infinitely small wavelength. The hard limit is 2 pixel to represent peak and through horizontal/vertically OR $\sqrt{2} \approx 1.4$ pixel diagonally. Thus C =2.6 is sensible results.

3. From the residuals graph, we can see it shows oscillatory behaviour around the fitted linear function. This is likely because of the discretised mode artifacts due to PBC(periodic boundary

condition). With PBC, we only can have a mode of

$$\lambda = \frac{N_{\text{pixel}}}{\sqrt{n_x^2 + n_y^2}} \tag{65}$$

where $N_{\text{pixel}}$ is the total number of pixel on one side which is 32 in this case. Thus in reality, the system can't theoretically have the continuous $\lambda_{\max}$. This sometimes makes double-peak power spectrum or skewed power spectrum. In either case, the symmetric Gaussian model for expected power spectrum isn't able to properly fit the data in. Therefore, this oscillatory artifacts can possibly arise in the process of fitting the Gaussian curve to the non gaussian data points.

Also the uncertainty and increases as $\lambda$ increase (smaller f). This is likely because the symmetric Gaussian model of expected power starts to fail as $f_{\max}$ gets closer to 0. Below is one of example. By eyes, it's able to catch the expected power spectrum is asymmetric and $f_{\max}$

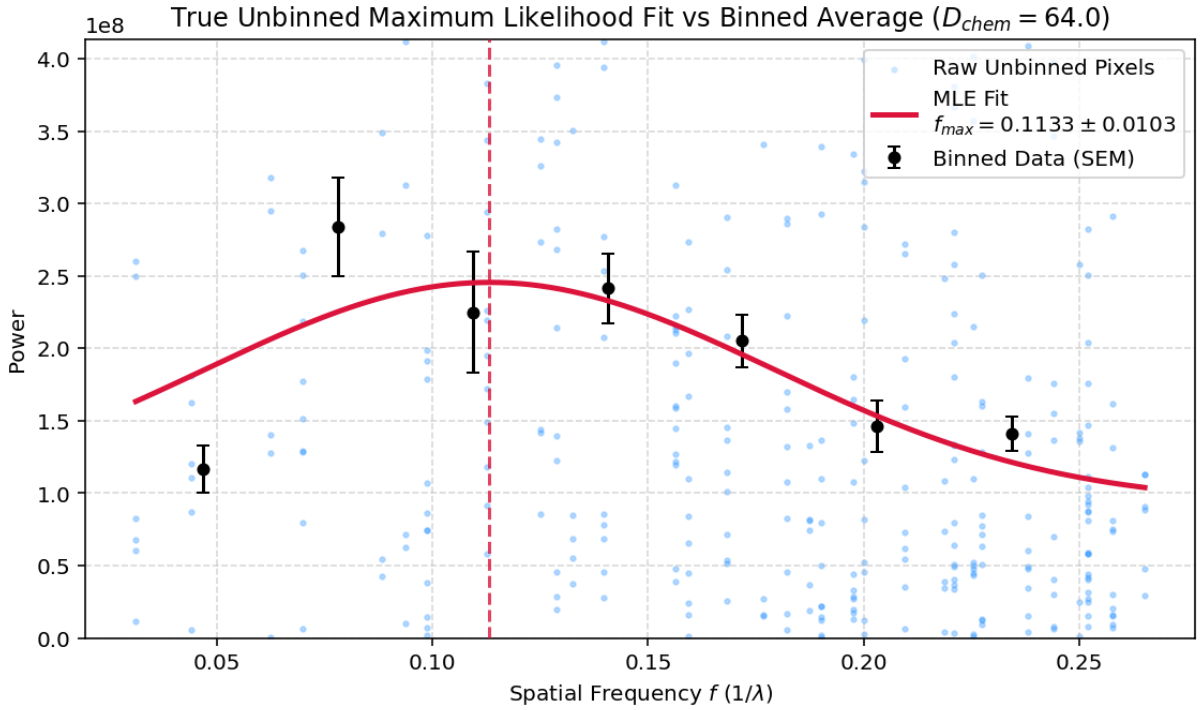



Figure 4: MLE fitting on asymmetric curve

must be smllaer than the current result. To address this problem, one possible way is using asymmetric distribution such as

$$S(f) = \frac{A}{f} \exp\left(-\frac{(\ln f - \mu)^2}{2w^2}\right) + C \tag{66}$$

This naturally accomodate asymmetric distribution.

# 4 Conclusion